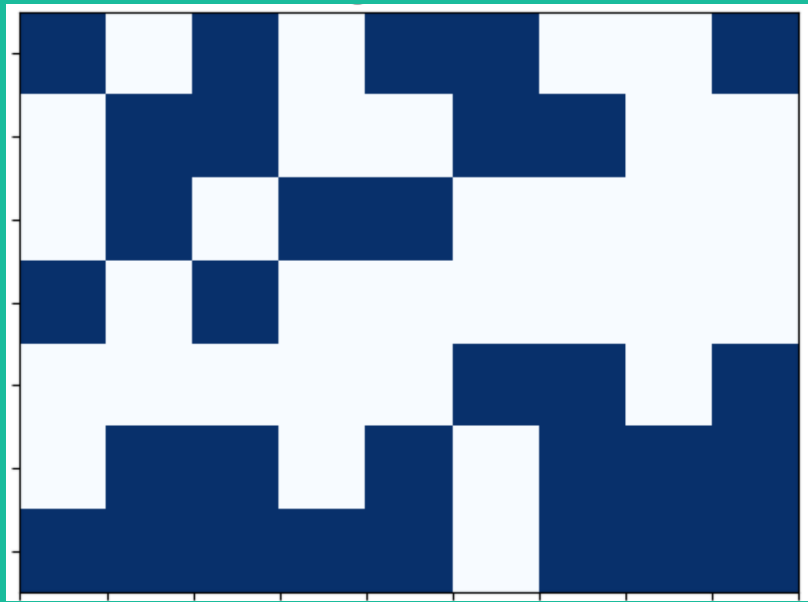




Final Assignment: Economic Complexity Report

Group 4

PART 1: Descriptive Statistics (1)

- Let's start with the input data

```
# Define matrix, connecting (imaginary) countries and activities.
```

```
countries=['Vale of Arryn','The North','The Riverlands', 'Dorne', 'The Stormland','The Westerlands','The Reach']  
activities=[f'P{j}' for j in range(9)]
```

```
matrix = [[1, 0, 1, 0, 1, 1, 0, 0, 1],  
          [0, 1, 1, 0, 0, 1, 1, 0, 0],  
          [0, 1, 0, 1, 1, 0, 0, 0, 0],  
          [1, 0, 1, 0, 0, 0, 0, 0, 0],  
          [0, 0, 0, 0, 0, 1, 1, 0, 1],  
          [0, 1, 1, 0, 1, 0, 1, 1, 1],  
          [1, 1, 1, 1, 1, 0, 1, 1, 1]]
```

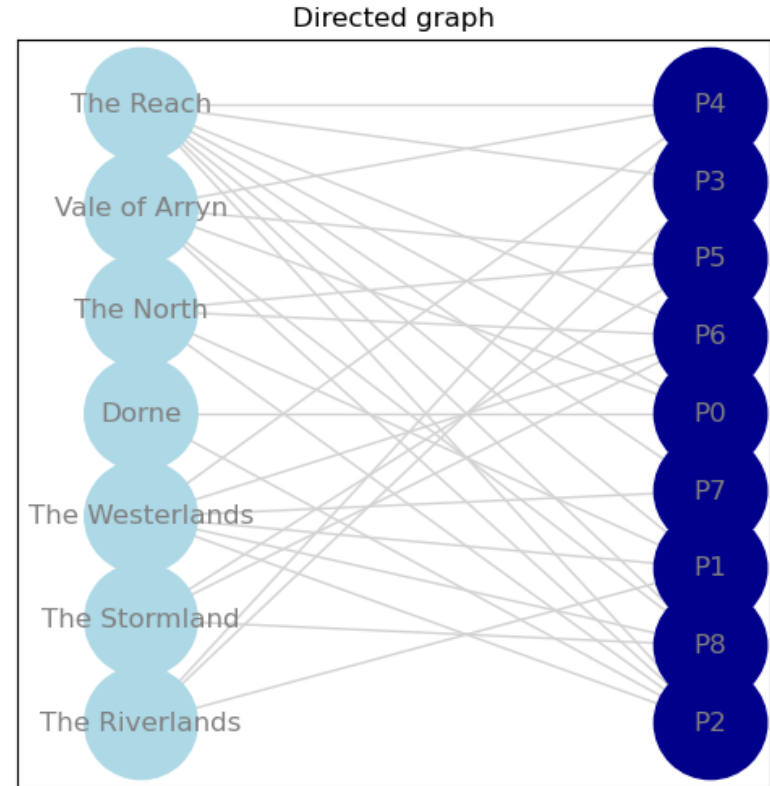
PART 1: Descriptive Statistics (2)

Original countries:

- index ' 0' - Vale of Arryn
- index ' 1' - The North
- index ' 2' - The Riverlands
- index ' 3' - Dorne
- index ' 4' - The Stormland
- index ' 5' - The Westerlands
- index ' 6' - The Reach

List of activities:

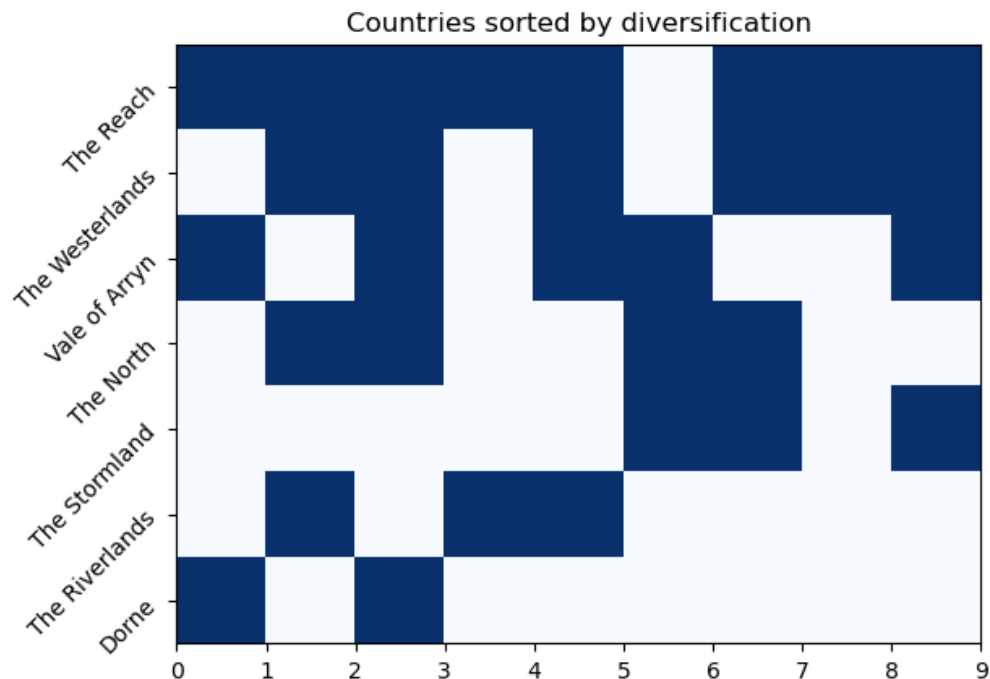
- index ' 0' - P0
- index ' 1' - P1
- index ' 2' - P2
- index ' 3' - P3
- index ' 4' - P4
- index ' 5' - P5
- index ' 6' - P6
- index ' 7' - P7
- index ' 8' - P8



PART 1: Descriptive Statistics (3)

Countries sorted by diversification (Most diverse at the top):

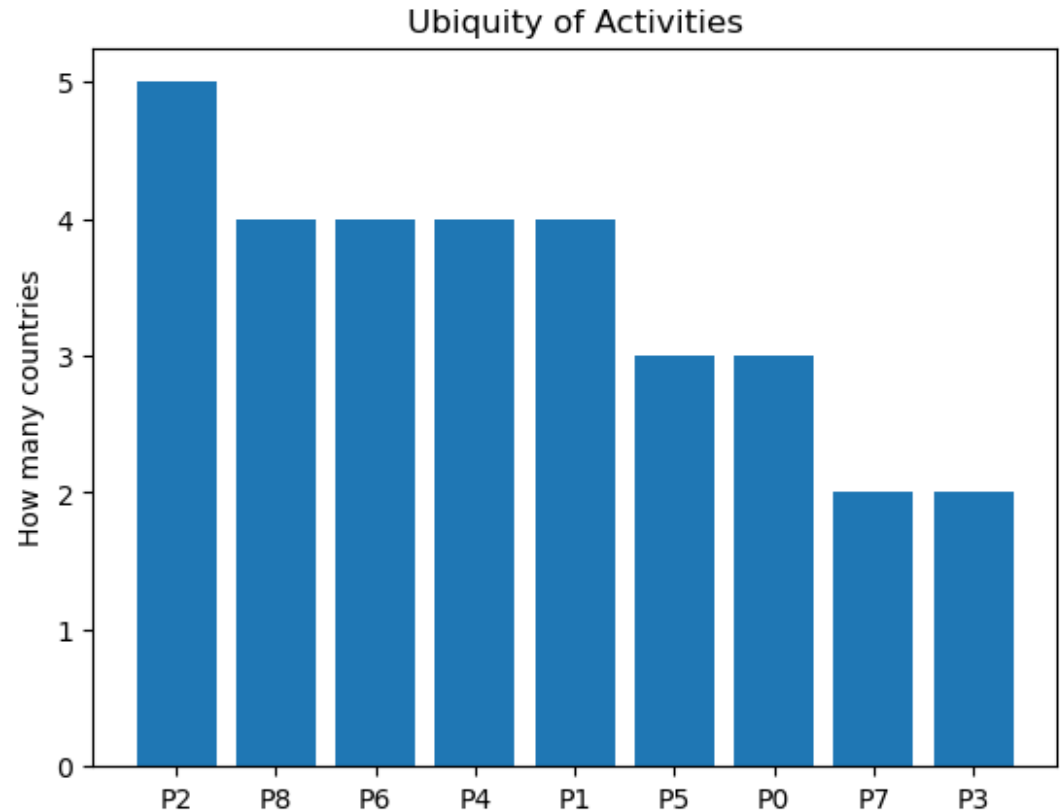
1. - The Reach (diversification: 8)
2. - The Westerlands (diversification: 6)
3. - Vale of Arryn (diversification: 5)
4. - The North (diversification: 4)
5. - The Stormland (diversification: 3)
6. - The Riverlands (diversification: 3)
7. - Dorne (diversification: 2)



PART 1: Descriptive Statistics (4)

- Ubiquity of Products / Activities
("Specialization"):

The most common product is **P2** as five countries (out of 7) are able to produce it; while **P7** and **P3** are the most rare to produce as only two countries are able to do so – either due to complexity or the product is too niche in other countries.



PART 2: Economic Complexity & Fitness Measures (1)

- ECI with eigenvalues

The ECI (Economic Complexity Index) was calculated using the eigenvalues and countries were reordered from the most developed to the least.

(There are for comparison also values with sign removed)

ECI as an eigenvalue problem:

0: Vale of Arryn	= -0.56136
1: The North	= -0.48151
2: The Riverlands	= 2.0548
3: Dorne	= -0.6958
4: The Stormland	= -1.1651
5: The Westerlands	= 0.30489
6: The Reach	= 0.54408

Ordered:

1. - (2) The Riverlands	= 2.0548
2. - (6) The Reach	= 0.54408
3. - (5) The Westerlands	= 0.30489
4. - (1) The North	= -0.48151
5. - (0) Vale of Arryn	= -0.56136
6. - (3) Dorne	= -0.6958
7. - (4) The Stormland	= -1.1651

Ordered (sign removed):

1. - (2) The Riverlands	= 2.0548
2. - (4) The Stormland	= 1.1651
3. - (3) Dorne	= 0.6958
4. - (0) Vale of Arryn	= 0.56136
5. - (6) The Reach	= 0.54408
6. - (1) The North	= 0.48151
7. - (5) The Westerlands	= 0.30489

PART 2: Economic Complexity & Fitness Measures (2)

- **PCI with eigenvalues**

The PCI (Product Complexity Index) was calculated using the eigenvalues and activities were reordered from the most sophisticated product to the least.

- Notice that **P3**, **P1** and **P4** are the most complex which are all produced by The Riverlands (first in ECI ranking).

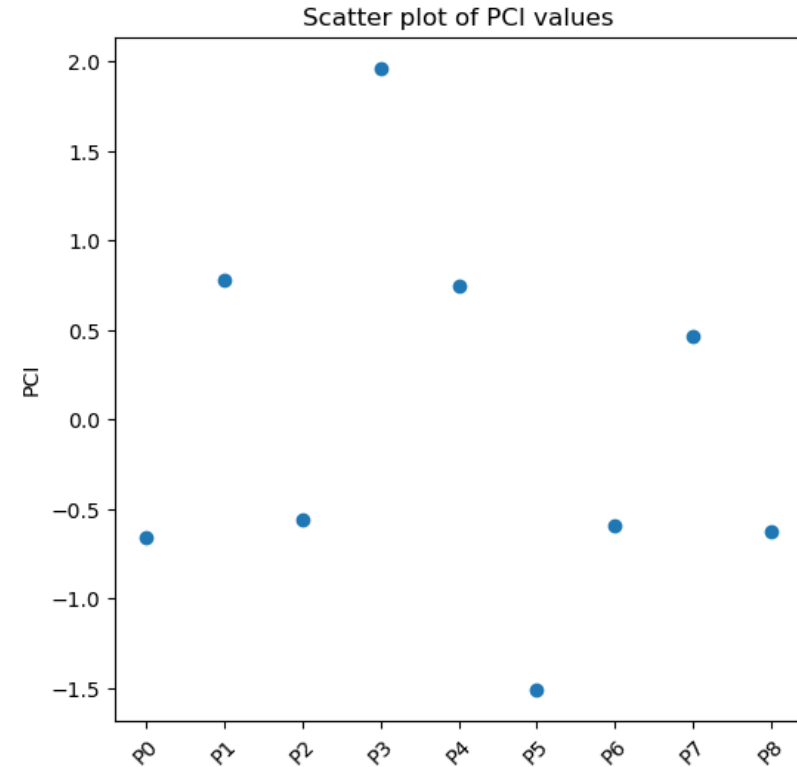
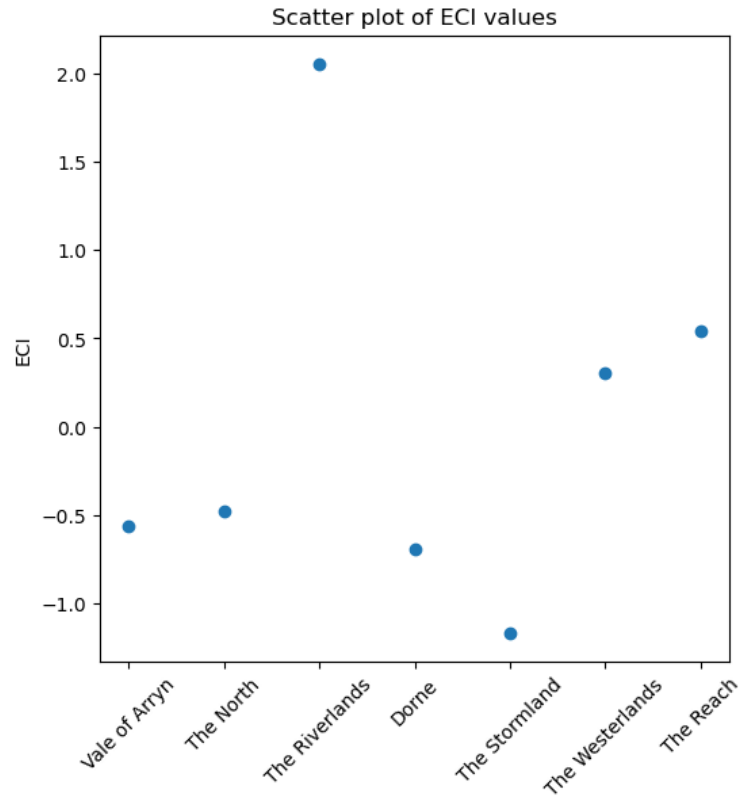
PCI as an eigenvalue problem:

0: P0 = -0.66008
1: P1 = 0.77784
2: P2 = -0.55819
3: P3 = 1.961
4: P4 = 0.7438
5: P5 = -1.5098
6: P6 = -0.59482
7: P7 = 0.46905
8: P8 = -0.62886

Ordered:

1. - (3) P3 = 1.961
2. - (1) P1 = 0.77784
3. - (4) P4 = 0.7438
4. - (7) P7 = 0.46905
5. - (2) P2 = -0.55819
6. - (6) P6 = -0.59482
7. - (8) P8 = -0.62886
8. - (0) P0 = -0.66008
9. - (5) P5 = -1.5098

PART 2: Economic Complexity & Fitness Measures (3)



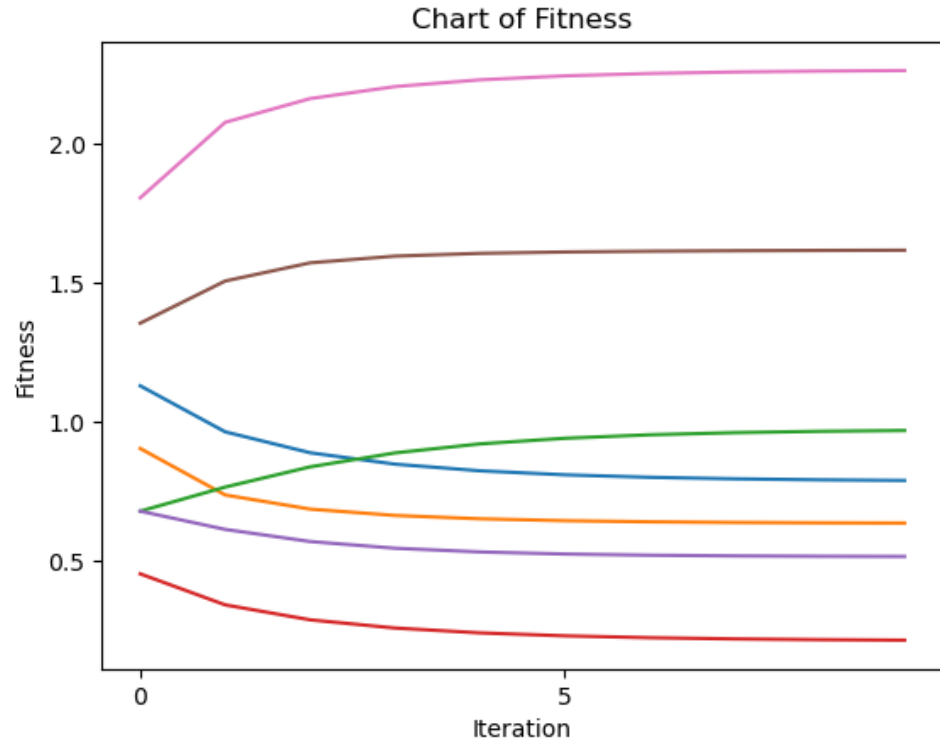
PART 2: Economic Complexity & Fitness Measures (4)

Fitness Ranking (Iterations: 10):

0: Vale of Arryn	= 0.78669
1: The North	= 0.63397
2: The Riverlands	= 0.97001
3: Dorne	= 0.21144
4: The Stormland	= 0.51372
5: The Westerlands	= 1.618
6: The Reach	= 2.2662

Ordered:

1. - (6) The Reach	= 2.2662
2. - (5) The Westerlands	= 1.618
3. - (2) The Riverlands	= 0.97001
4. - (0) Vale of Arryn	= 0.78669
5. - (1) The North	= 0.63397
6. - (4) The Stormland	= 0.51372
7. - (3) Dorne	= 0.21144



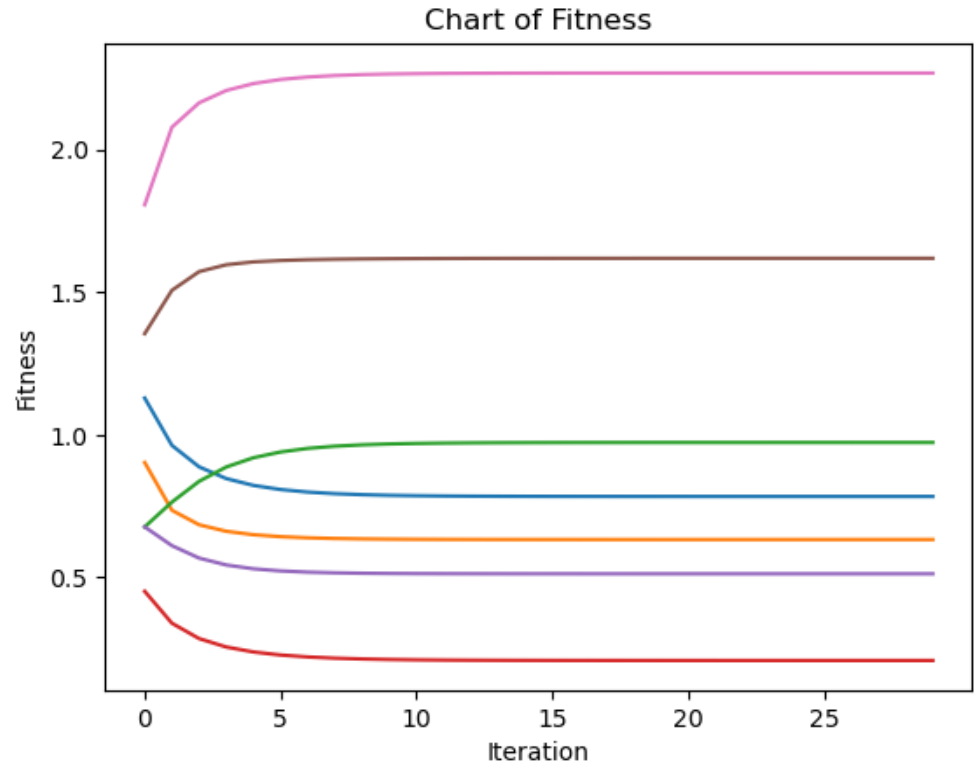
PART 2: Economic Complexity & Fitness Measures (5)

Fitness Ranking (Iterations: 30):

0: Vale of Arryn	= 0.78405
1: The North	= 0.6331
2: The Riverlands	= 0.97329
3: Dorne	= 0.20917
4: The Stormland	= 0.51304
5: The Westerlands	= 1.619
6: The Reach	= 2.2684

Ordered:

1. - (6) The Reach	= 2.2684
2. - (5) The Westerlands	= 1.619
3. - (2) The Riverlands	= 0.97329
4. - (0) Vale of Arryn	= 0.78405
5. - (1) The North	= 0.6331
6. - (4) The Stormland	= 0.51304
7. - (3) Dorne	= 0.20917



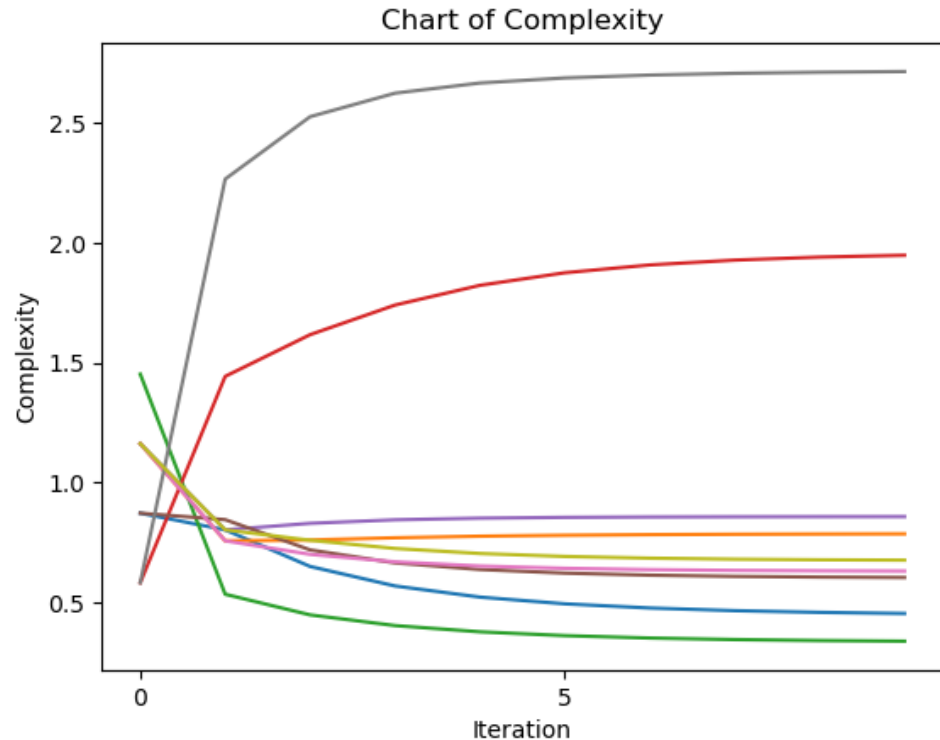
PART 2: Economic Complexity & Fitness Measures (6)

Complexity Ranking (Iterations: 10):

0: P0 = 0.44905
1: P1 = 0.78499
2: P2 = 0.33458
3: P3 = 1.9532
4: P4 = 0.85673
5: P5 = 0.60127
6: P6 = 0.62871
7: P7 = 2.7175
8: P8 = 0.67391

Ordered:

1. - (7) P7 = 2.7175
2. - (3) P3 = 1.9532
3. - (4) P4 = 0.85673
4. - (1) P1 = 0.78499
5. - (8) P8 = 0.67391
6. - (6) P6 = 0.62871
7. - (5) P5 = 0.60127
8. - (0) P0 = 0.44905
9. - (2) P2 = 0.33458



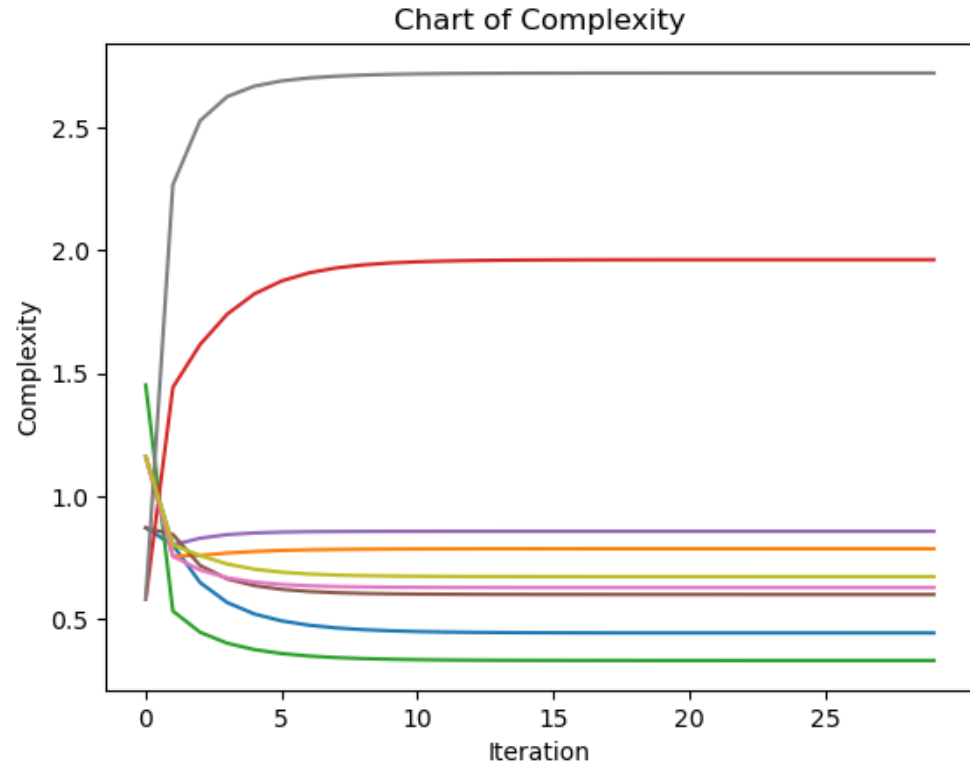
PART 2: Economic Complexity & Fitness Measures (7)

Complexity Ranking (Iterations: 30):

0: P0 = 0.44333
1: P1 = 0.78579
2: P2 = 0.33129
3: P3 = 1.9617
4: P4 = 0.85688
5: P5 = 0.59955
6: P6 = 0.6279
7: P7 = 2.7211
8: P8 = 0.67248

Ordered:

1. - (7) P7 = 2.7211
2. - (3) P3 = 1.9617
3. - (4) P4 = 0.85688
4. - (1) P1 = 0.78579
5. - (8) P8 = 0.67248
6. - (6) P6 = 0.6279
7. - (5) P5 = 0.59955
8. - (0) P0 = 0.44333
9. - (2) P2 = 0.33129



PART 2: Economic Complexity & Fitness Measures (8)

	Country	Diversification	ECI	Fitness	ECI Ranking	Fitness Ranking	Difference
1	The Riverlands	3	2.054840	0.973252	1	3	-2
2	The Reach	8	0.544078	2.268360	2	1	1
3	The Westerlands	6	0.304887	1.618939	3	2	1
4	The North	4	-0.481505	0.633110	4	5	-1
5	Vale of Arryn	5	-0.561359	0.784084	5	4	1
6	Dorne	2	-0.695802	0.209205	6	7	-1
7	The Stormland	3	-1.165138	0.513049	7	6	1

PART 2: Economic Complexity & Fitness Measures (9)

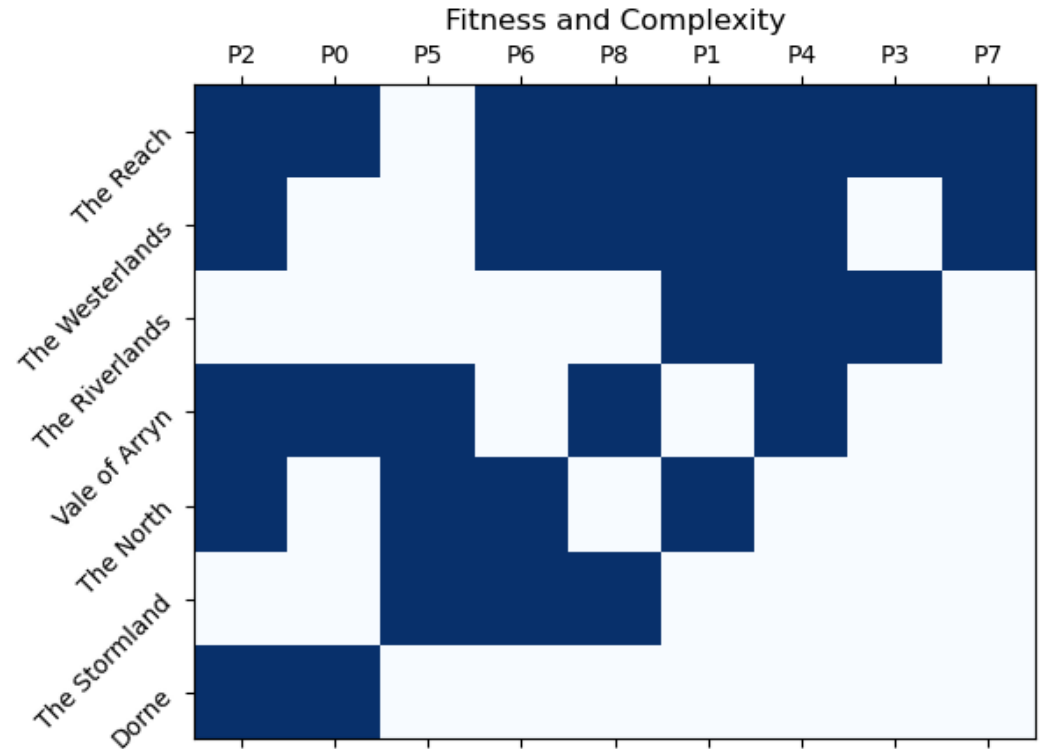
	Activity	Ubiquity	PCI	Complexity	PCI Ranking	Complexity Ranking	Difference	
1	P3	2	1.961043	1.961588	1	2	-1	
2	P1	4	0.777844	0.785781	2	4	-2	
3	P4	4	0.743803	0.856881	3	3	0	
4	P7	2	0.469049	2.721019	4	1	3	
5	P2	5	-0.558192	0.331338	5	9	-4	!
6	P6	4	-0.594818	0.627909	6	6	0	
7	P8	4	-0.628859	0.672498	7	5	2	
8	P0	3	-0.660084	0.443413	8	8	0	*
9	P5	3	-1.509787	0.599571	9	7	2	



PART 2: Economic Complexity & Fitness Measures (10)

- Fitness and Complexity (1)

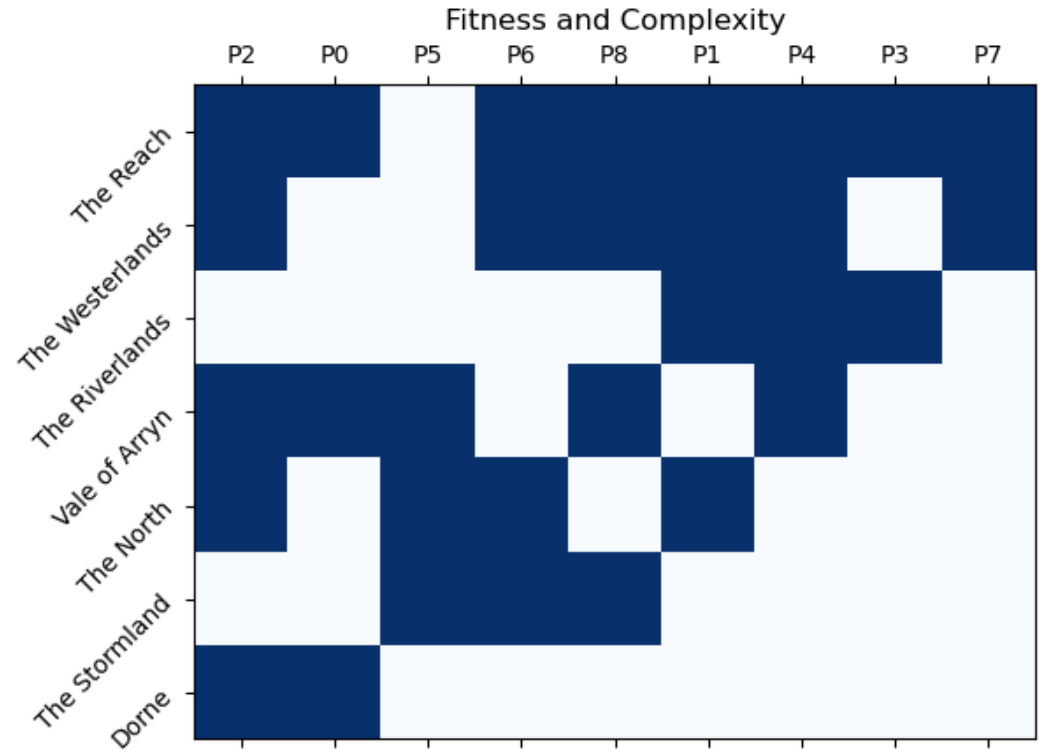
This map shows that The Reach and The Westerlands are the most developed economies with the widest activities (production). Also we can see that the **P7** and **P3** are the most rare products to make – as only two countries are able or chose to produce them.



PART 2: Economic Complexity & Fitness Measures (11)

- Fitness and Complexity (2)

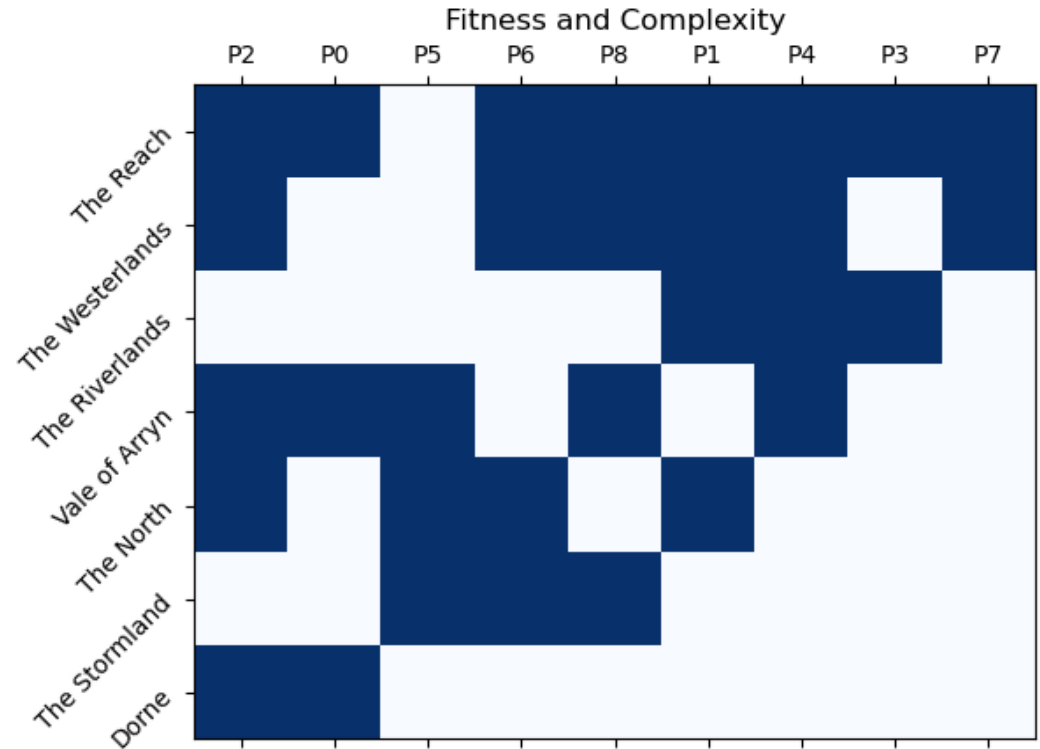
The Riverlands seems to specialize in top-end products of **P1**, **P4** and **P3** (in both PCI and Complexity) while country of Dorne is limited economy targeting the simplest of products **P0** and the most ubiquitous **P2** (mid in PCI ranking but bottom in Complexity).



PART 2: Economic Complexity & Fitness Measures (12)

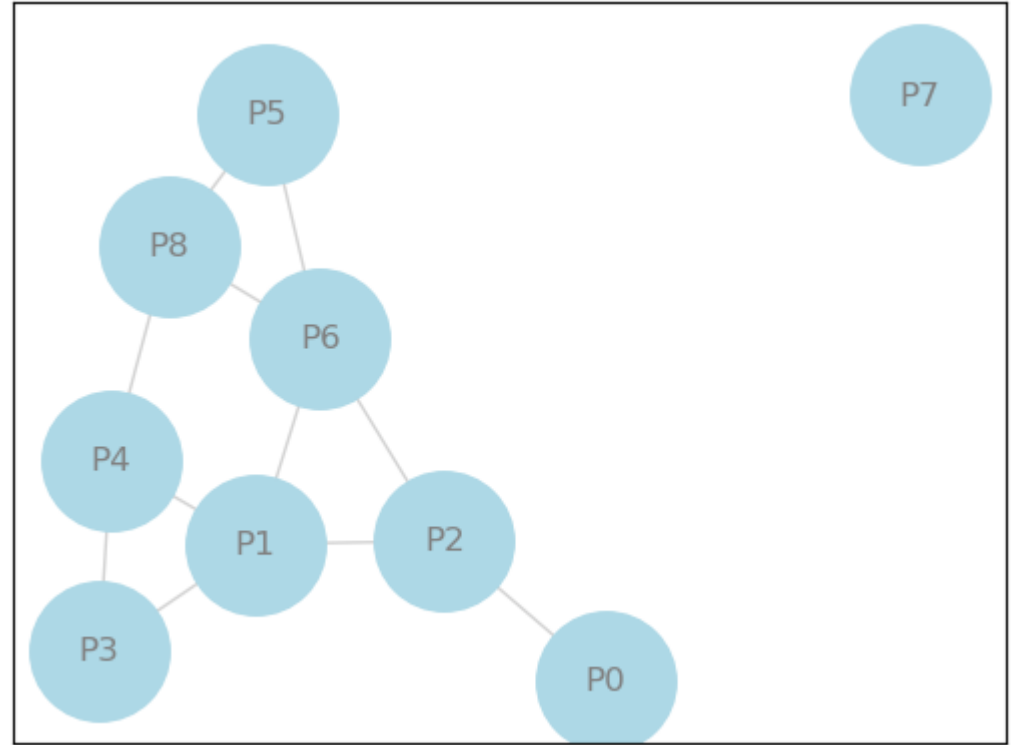
- Fitness and Complexity (3)

The Stormland is another low-tier economy competing with the Dorne for the last developed spot in both ECI and Fitness ranking. It produces another cluster of the most simplest of products: **P5**, **P6** and **P8**



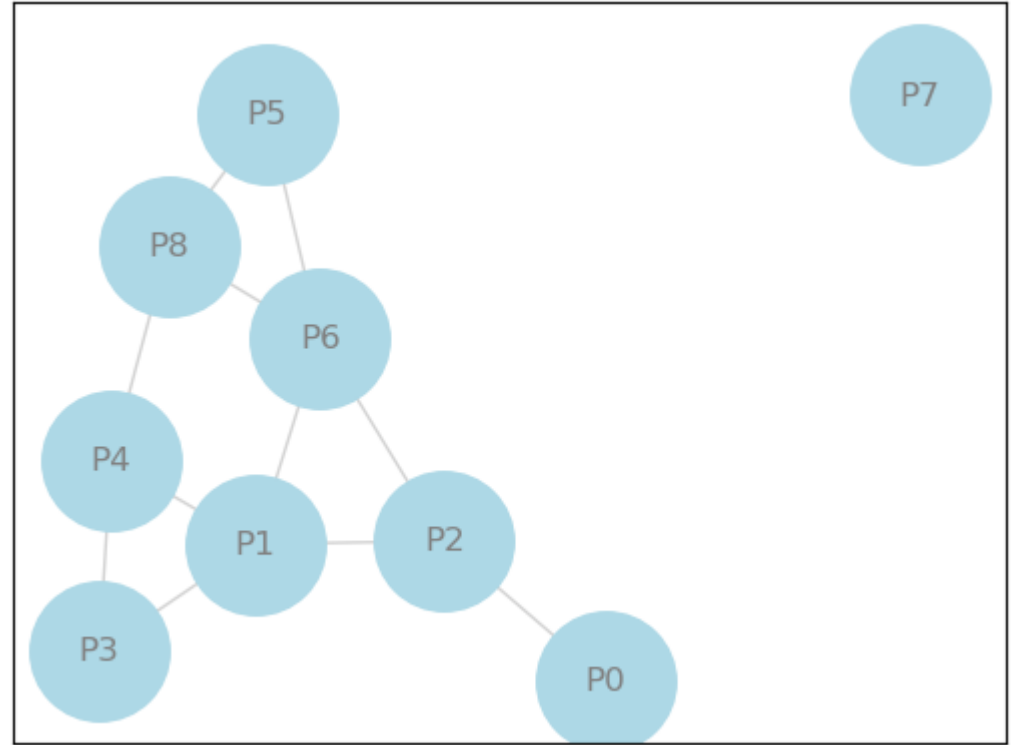
PART 3: Relatedness Between Activities (1)

- We can see that product **P7** is isolated from the rest of the network and it is the least ubiquitous product to make – **only two countries developed capabilities to produce it**. It is number one in Complexity ranking.



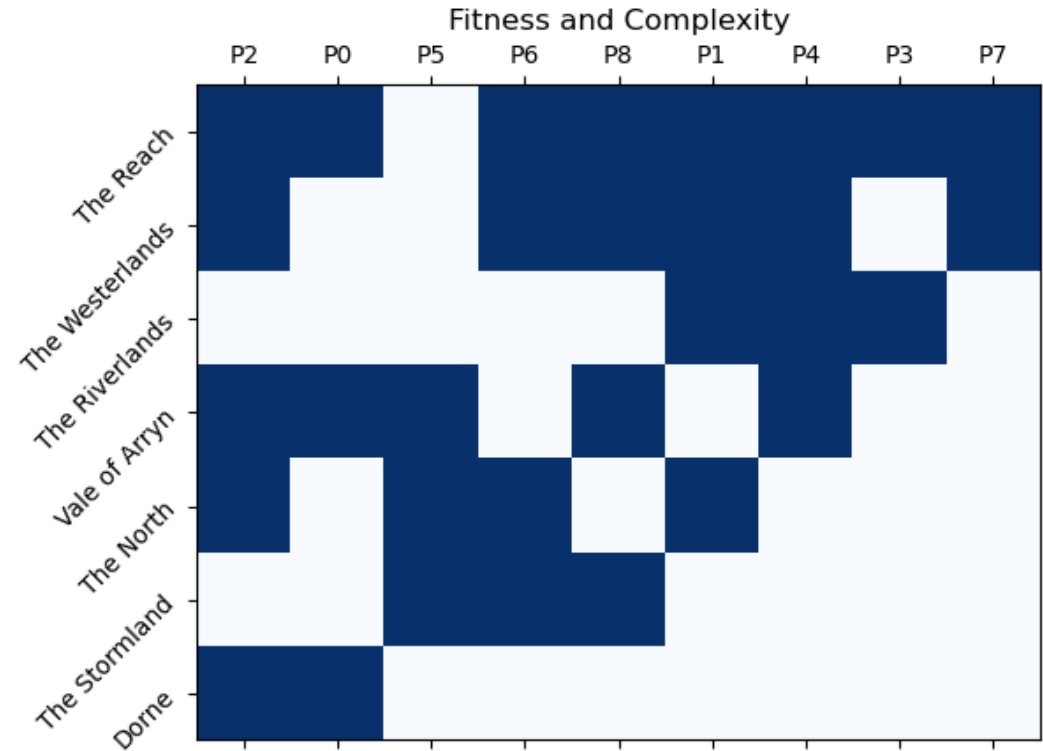
PART 3: Relatedness Between Activities (2)

- P2 is the most ubiquitous product while capability P1 is an entry point to the most sophisticated of activities (except P7).
- Every country having capability of producing P2 has capability of producing P0 (next slide).

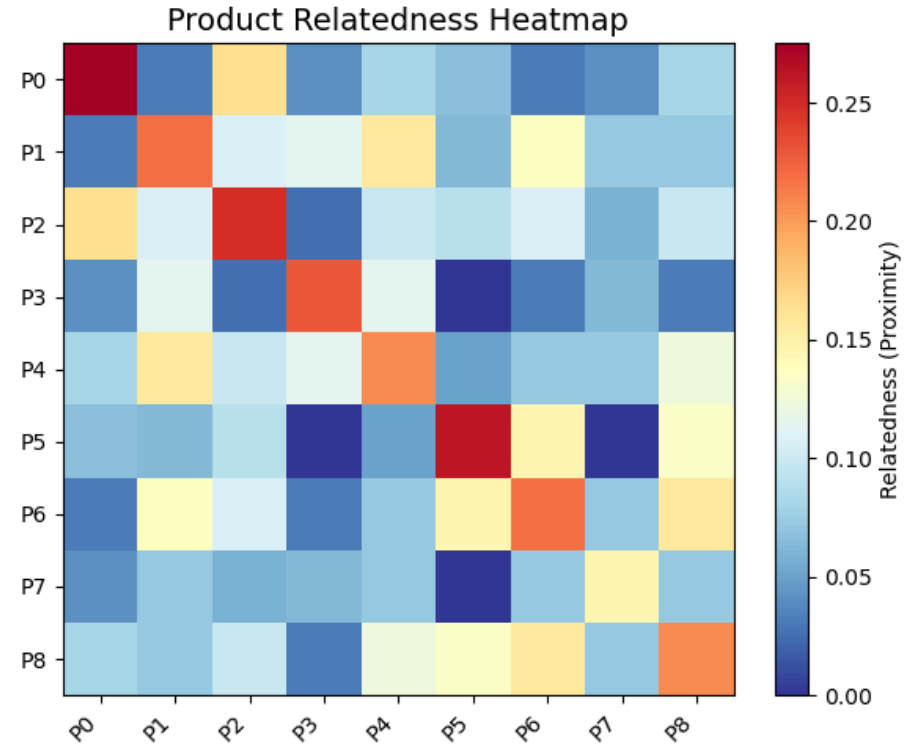
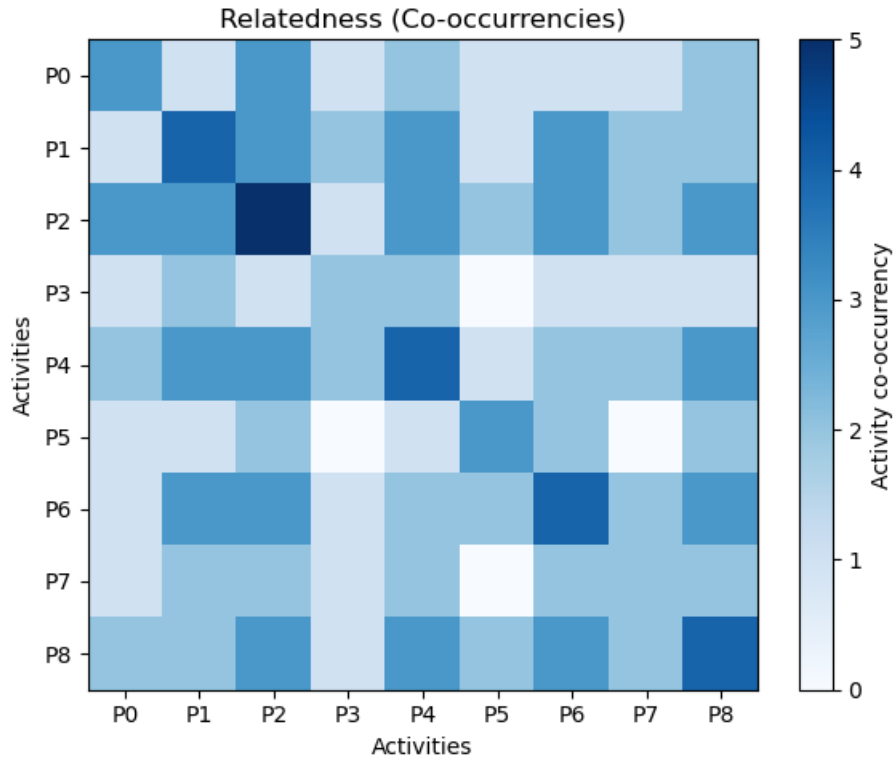


PART 3: Relatedness Between Activities (3)

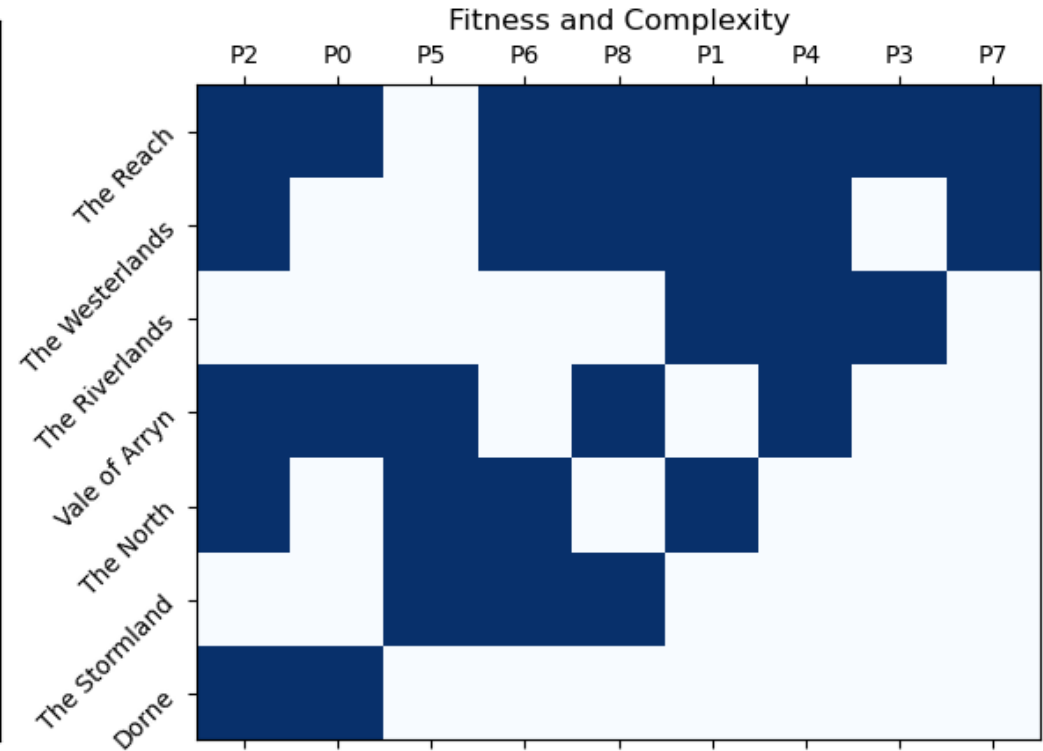
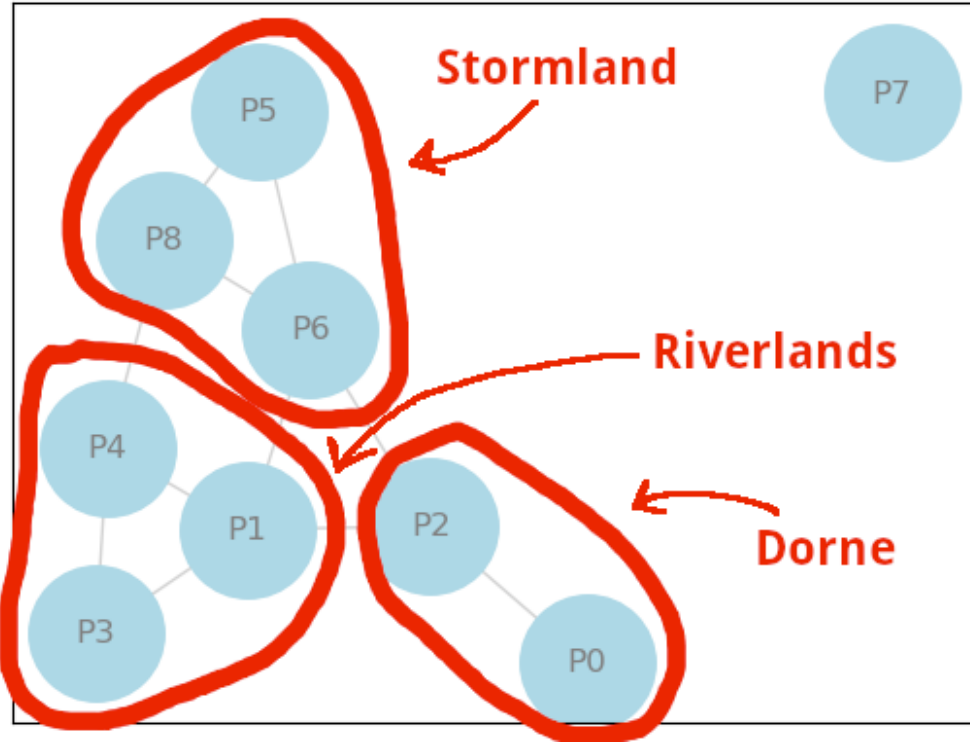
- Here we can confirm that every country producing P0 is also producing P2.
- We can also argue that P1 differentiate the most advanced economies from the rest. P7 is arguably the most difficult capability to obtain.



PART 3: Relatedness Between Activities (4)

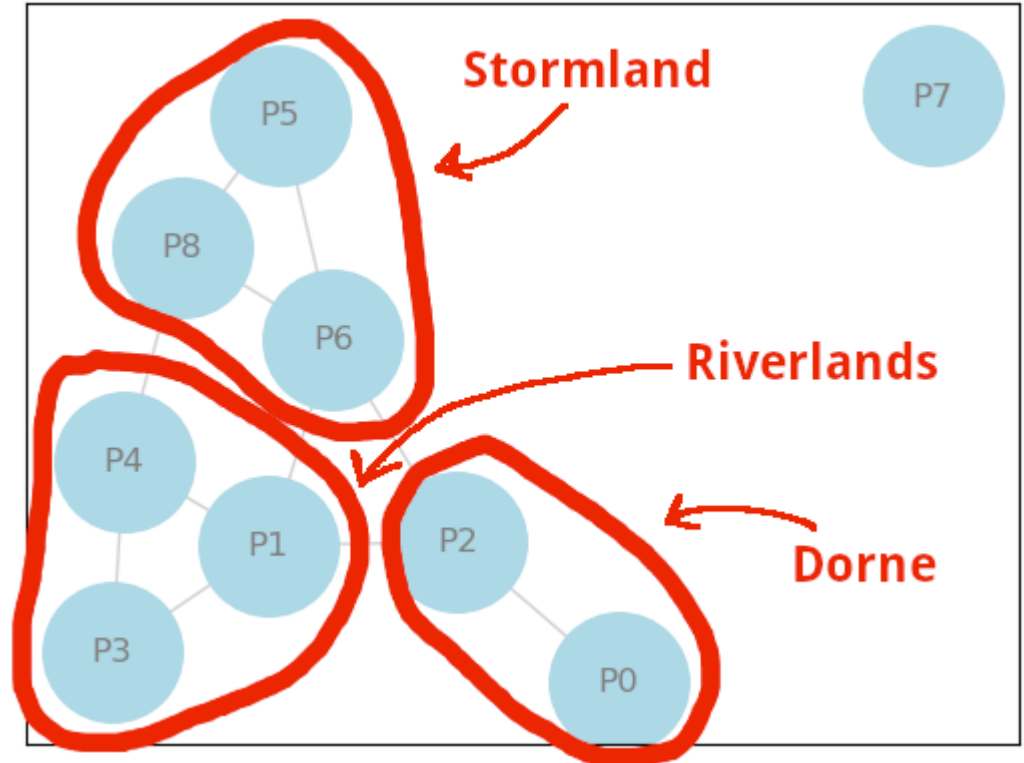


PART 4: Diversification Strategies (1)



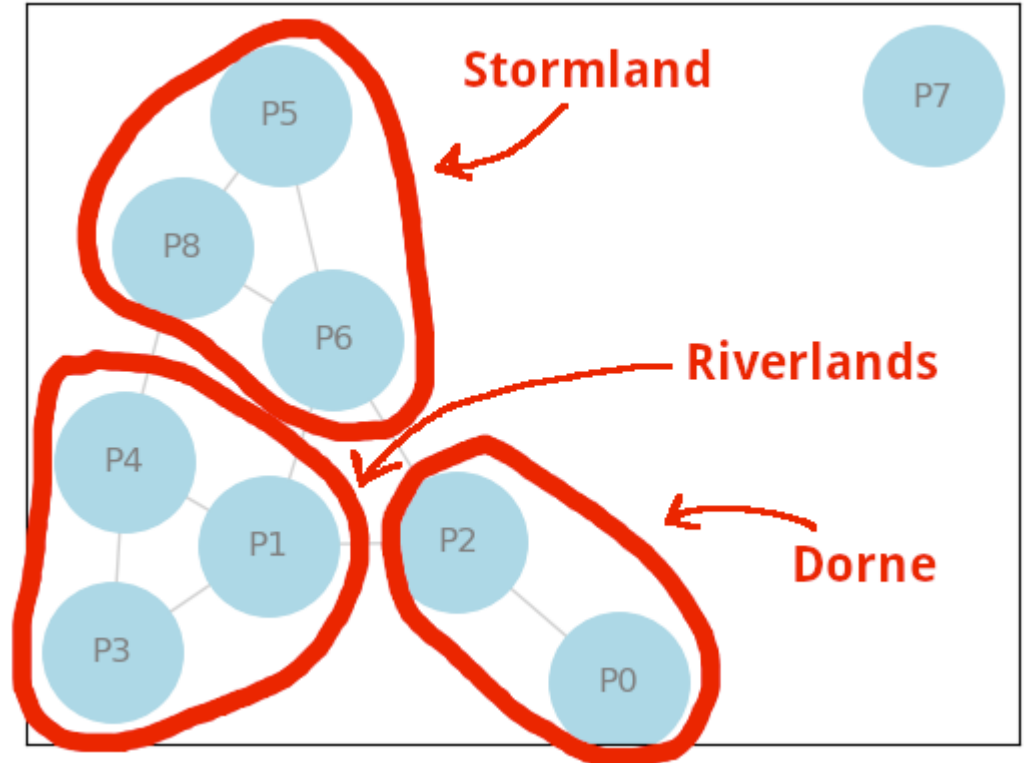
PART 4: Diversification Strategies (2)

- There is a reasonable path for **Dorne** to diversify their economy by trying to develop capability to produce product **P6** as that might open door for even more activities – albeit in the low-tier only.



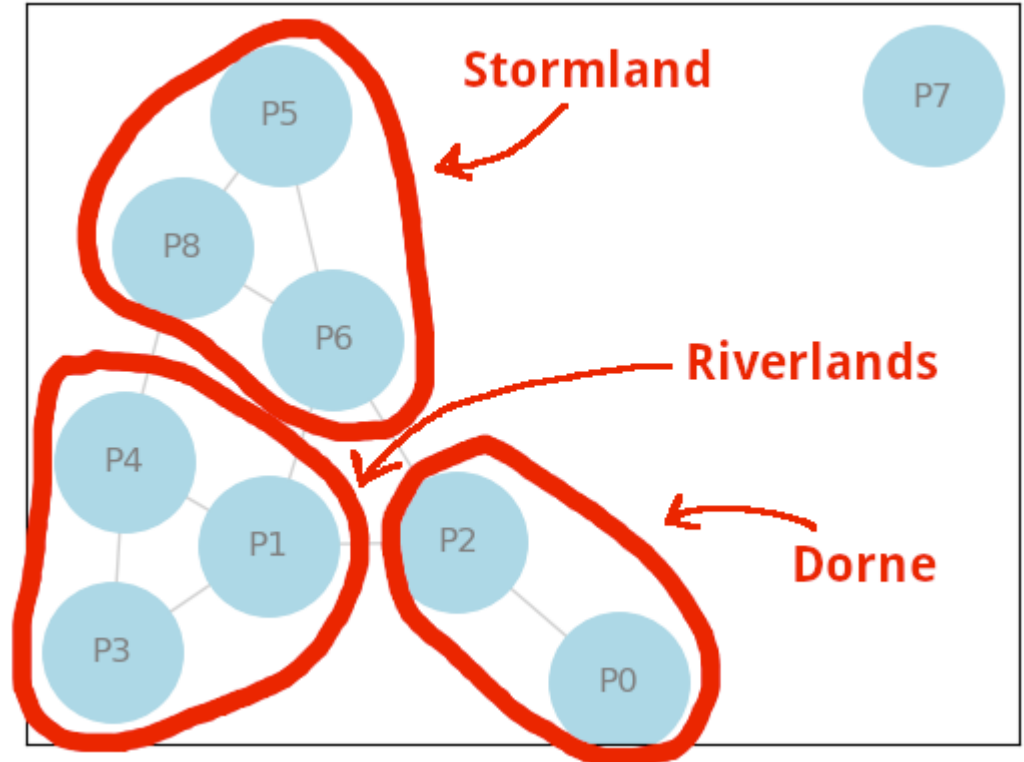
PART 4: Diversification Strategies (3)

- Stormland could form from the opposite direction extend their base by adopting capabilities of **P2** and **P0** – both Dorne and Stormland are undeniably the least developed economies here.



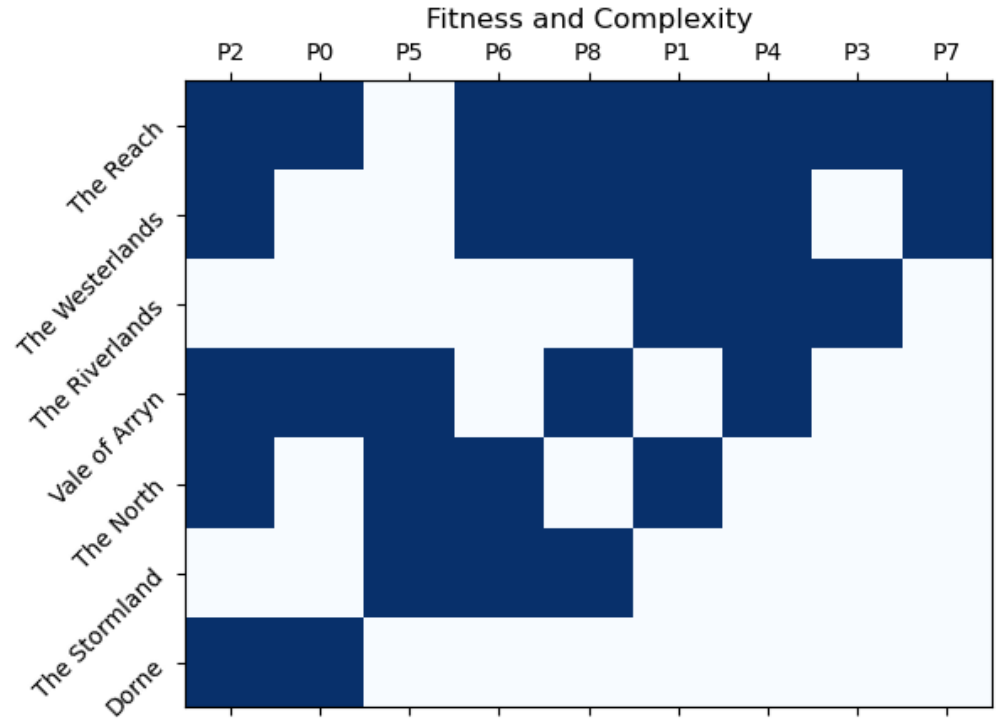
PART 4: Diversification Strategies (4)

- On the other hand developing capability P6 might help The Riverlands to access the more ubiquitous and less complex products of the lower segments.



PART 4: Diversification Strategies (5)

- The North and Vale of Arryn are already positioned well to try develop the top-end capabilities of P4, P3 and P1, P3 respectively.



PART 5: Workflow and Contributions

- This slides and Jupyter (python) notebook is hosted on github:
 - https://github.com/ospalax/unu_merit_eci
- The collaboration started via initial email and then moved to a google's hangout chat room for offline async communication:
 - <https://chat.google.com/app/chat/AAQAdfTizyY>
 - Every group member had a gmail account and was invited



Thank You